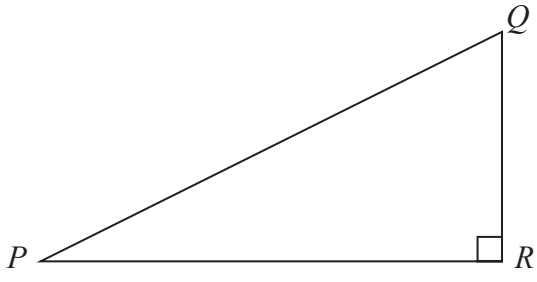
12 Here is triangle PQR.



The length of QR is 60% of the length of PR.

Find the value of sin QPR.
Give your answer correct to 3 significant figures.

Take PR=5 units and QR=3 units (60%=3/5).

1 mark
Equivalent side ratio

$PQ = \sqrt{(5^2 + 3^2)} = \sqrt{34}$ units.

1 mark
Find hypotenuse

$$\sin QPR = \frac{3}{\sqrt{34}} = 0.514495\ldots \approx 0.514$$

1 mark
0.514 to3sf

(Total for Question 12 is 3 marks)